Sonning kvadrat ildizi va kub ildizi

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. $\sqrt{3^4 \cdot 5^2 \cdot 7^2}$ **ni toping.**

A) 1 575 **B)** 945 **C)** 315 **D)** 105

2. $\sqrt[3]{2^9 \cdot 5^6}$ **ni toping.**

A) 200 **B)** 25 **C)** 1 000 **D)** 40

3. $126 \cdot n$ **to'la kvadrat.** n **ning eng kichik natural qiymatini toping.**

A) 14 **B)** 126 **C)** 2 **D)** 7

4. $72 \cdot m$ **to'la kub.** m **ning eng kichik natural qiymatini toping.**

A) 9 **B)** 72 **C)** 3 **D)** 2

5. **Rasmda** $180 \cdot m$ **to'la kub bo'lishi uchun ko'paytuvchilar uchta guruhga joylangan; punktir doirachalar — yetishmagan ko'paytuvchilar.** m **ning eng kichik natural qiymati nechaga teng?**



A) 30 **B)** 5
C) 150 **D)** 900

6. $\frac{1800}{a}$ **to'la kvadrat.** a **ning eng kichik natural qiymatini toping.**

A) 5 **B)** 2 **C)** 8 **D)** 3

7. **1 512 ning bo'luvchilari ichida to'la kvadrat bo'ladigan eng kattasini toping.**

A) 9 **B)** 216 **C)** 36 **D)** 144

8. **3 456 ning bo'luvchilari ichida to'la kub bo'ladigan eng kattasini toping.**

A) 64 **B)** 576 **C)** 216 **D)** 1 728

9. $9 \cdot n$ **ham to'la kvadrat, ham to'la kub.** n **ning eng kichik natural qiymatini toping.**

A) 27 **B)** 3 **C)** 9 **D)** 81

10. $2^3 \cdot 7^3$ **ning kvadratini toping, so'ng natijaning kub ildizini toping.**

A) $2^6 \cdot 7^6$ **B)** $2 \cdot 7$ **C)** $2^3 \cdot 7^3$ **D)** $2^2 \cdot 7^2$

11. **16 va 36 ning EKUKining kvadrat ildizini toping.**

A) 72 **B)** 12 **C)** 24 **D)** 144

12. **1 080 va 1 512 ning EKUBining kub ildizini toping.**

A) 72 **B)** 216 **C)** 6 **D)** 36

13. **Kub shaklidagi akvariumning hajmi 6 859 dm³. Uning barcha qirralari uzunliklarining yig'indisi qancha? (Kalkulyatordan foydalanish mumkin.)**

A) 76 dm **B)** 228 dm **C)** 19 dm **D)** 114 dm

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14. **Kvadrat shaklidagi stadion maydonining yuzi 2 704 m². Sportchi maydon chetidan bir marta aylanib yugurdi. U necha metr yugurdi?**

A) 52 m **B)** 208 m **C)** 676 m **D)** 104 m

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15. **Rasmda 1 400 ning tub ko'paytuvchilari ikkita bir xil guruhga ajratilgan, ba'zilari ortib qoldi.** $1400 \cdot n$ **to'la kvadrat bo'lishi uchun** n **ning eng kichik natural qiymati nechaga teng?**



A) 7 **B)** 2

C) 1 400 **D)** 14

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16. **□ — noma'lum raqam.** $3\square64$ **soni to'la kvadrat. □ nechaga teng?**

A) 3 **B)** 6 **C)** 8 **D)** 1

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17. $y = 3 \cdot 5^3$ **va** $z = 3^3 \cdot 5$**.** $y \cdot z$ **ko'paytmaning kvadrat ildizini toping.**

A) 15 **B)** 75 **C)** 225 **D)** 50 625

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18. **Qaysi son ham to'la kvadrat, ham to'la kub?**

A) 343 **B)** 216 **C)** 100 **D)** 729

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19. **Aziz shunday yechdi: «**$\sqrt[3]{3^6 \cdot 7^3} = 3^3 \cdot 7 = 189$**». Uning xatosi nimada?**

A) Xato yo'q

B) Darajani 3 ga emas, 2 ga bo'lgan; javob 63

C) Javob $3 \cdot 7 = 21$

D) 7 ni yozmasligi kerak edi; javob 27

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20. **Qaysi sonning kvadrat ildizi 12 bilan 13 orasida emas?**

A) 170 **B)** 168 **C)** 150 **D)** 160

Javoblar

Sonning kvadrat ildizi va kub ildizi · 2-test (qiyinroq)

№	Javob	Qiymati
1	**C**	315
2	**A**	200
3	**A**	14
4	**C**	3
5	**C**	150
6	**B**	2
7	**C**	36
8	**D**	1 728
9	**D**	81
10	**D**	$2^2 \cdot 7^2$

№	Javob	Qiymati
11	**B**	12
12	**C**	6
13	**B**	228 dm
14	**B**	208 m
15	**D**	14
16	**A**	3
17	**C**	225
18	**D**	729
19	**B**	Darajani 3 ga emas, 2 ga bo'lgan; javob 63
20	**A**	170

Qisqa yechimlar

1. Har bir daraja 2 marta kichrayadi: $3^2 \cdot 5 \cdot 7 = 9 \cdot 35$ = **315**.
(105 = 3 · 5 · 7 — 3 ning darajasi 4 marta kamaytirilgan.)

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2. Kub ildizda darajalar 3 marta kichrayadi: $2^3 \cdot 5^2 = 8 \cdot 25$ = **200**.
(40 = $2^3 \cdot 5$ — 5 ning darajasi xato.)

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3. $126 = 2 \cdot 3^2 \cdot 7$: 3^2 ikkita guruhga bo'linadi, 2 va 7 ortib qoladi.
Ikkinchi guruhga 2 va 7 yetishmaydi: $n = 2 \cdot 7$ = **14**.
Tekshirish: $126 \cdot 14 = 1764 = 42^2$. (126 ham mos, lekin eng kichigi emas.)

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4. $72 = 2^3 \cdot 3^2$: 2^3 uchta guruhga bittadan tushadi, uchinchi guruhga 3 yetishmaydi.
m = **3**: $72 \cdot 3 = 216 = 6^3$.
(2 — to'la kvadrat uchun javob: $144 = 12^2$.)

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5. $180 = 2^2 \cdot 3^2 \cdot 5$, har guruhda 2, 3, 5 bo'lishi kerak.
Yetishmaydi: bitta 2, bitta 3 va ikkita 5. $m = 2 \cdot 3 \cdot 5^2$ = **150**.
Tekshirish: $180 \cdot 150 = 27000 = 30^3$.

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6. $1800 = 2^3 \cdot 3^2 \cdot 5^2$: faqat 2 ning darajasi toq — bitta 2 ortib qoladi.
Uni bo'lib olib tashlaymiz: a = **2**, $1800 : 2 = 900 = 30^2$.

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7. $1512 = 2^3 \cdot 3^3 \cdot 7$. Har bir tub sondan juft darajaning eng kattasini olamiz: $2^2 \cdot 3^2$ = **36**.
216 to'la kub, to'la kvadrat emas; 144 esa 1 512 ning bo'luvchisi emas.

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8. $3456 = 2^7 \cdot 3^3$. 3 ga bo'linadigan eng katta darajalar: 2^6 va 3^3.
$2^6 \cdot 3^3$ = **1 728** $= 12^3$.
576 = 24^2 — to'la kvadrat, lekin to'la kub emas.

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9. $9 = 3^2$. Daraja 2 ga ham, 3 ga ham, ya'ni 6 ga bo'linishi kerak: $3^6 = 9 \cdot 81$.
n = **81**: $729 = 27^2 = 9^3$.
(3 — faqat kub uchun, 9 — faqat kvadrat uchun.)

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10. Kvadrati: $(2^3 \cdot 7^3) \cdot (2^3 \cdot 7^3) = 2^6 \cdot 7^6$.
Kub ildiz — darajalar 3 marta kichrayadi: $2^2 \cdot 7^2$ = 196.

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11. $16 = 2^4$, $36 = 2^2 \cdot 3^2$, EKUK $= 2^4 \cdot 3^2 = 144$.
$\sqrt{144}$ = **12**. (72 — 144 ni 2 ga bo'lish.)

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12. $1080 = 2^3 \cdot 3^3 \cdot 5$, $1512 = 2^3 \cdot 3^3 \cdot 7$, EKUB $= 2^3 \cdot 3^3 = 216$.
$\sqrt[3]{216} = 2 \cdot 3$ = **6**. (72 — 216 ni 3 ga bo'lish.)

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13. Kalkulyatorda $\sqrt[3]{6859} = 19$: qirrasi 19 dm ($19 \cdot 19 \cdot 19 = 6859$).
Kubning 12 ta qirrasi bor: $12 \cdot 19$ = **228 dm**.
(76 dm — kvadrat perimetri kabi 4 ga ko'paytirilgan.)

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14. $2704 = 2^4 \cdot 13^2$, tomoni $2^2 \cdot 13 = 52$ m.
Perimetr: $4 \cdot 52$ = **208 m**.
(676 — yuzni 4 ga bo'lish xatosi.)

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15. Ortib qolgan ko'paytuvchilar: 2 va 7. Har biriga jufti kerak.
$n = 2 \cdot 7$ = **14**: $1400 \cdot 14 = 19600 = 140^2$.
(1 400 ham mos, lekin eng kichigi emas.)

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16. 3 064 dan 3 964 gacha: $56^2 = 3136$, $57^2 = 3249$, $58^2 = 3364$, …, $62^2 = 3844$.
Oxiri 64 bo'lgani faqat $3364 = 58^2$. □ = **3**.

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17. $y \cdot z = 3^4 \cdot 5^4$ — darajalar juft.
$\sqrt{y \cdot z} = 3^2 \cdot 5^2$ = **225**. (50 625 — bu $y \cdot z$ ning o'zi, 15 — darajalar 4 marta kamaytirilgan.)

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18. $729 = 3^6$: 6 son 2 ga ham, 3 ga ham bo'linadi — $27^2 = 9^3$. Javob: **729**.
343 va 216 faqat to'la kub, 100 faqat to'la kvadrat.

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19. Kub ildizda daraja ko'rsatkichlari 3 ga bo'linadi: $3^6 \to 3^2$, $7^3 \to 7$.
To'g'risi: $3^2 \cdot 7 = 63$. Aziz **darajani 2 ga bo'lgan**, ya'ni kvadrat ildiz olgandek ishlagan.

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20. $12^2 = 144$, $13^2 = 169$. Ildizi 12 bilan 13 orasidagi sonlar: 145 dan 168 gacha.
170 > 169, demak $\sqrt{170}$ 13 dan katta.
